

# MySQL Transaction Control: COMMIT, ROLLBACK, SAVEPOINT

## Hands-On Learning Guide for Corporate SQL Training

MySQL Transaction Control Language (TCL) commands are used to manage transactions safely. These commands ensure data consistency and reliability in enterprise applications. Transactions are heavily used in: - Banking Systems - Payment Gateways - E-Commerce Applications - Airline Reservation Systems - ERP Software

## What is a Transaction?

A transaction is a group of SQL statements executed as one unit. Example: When money is transferred from one bank account to another: 1. Money is deducted from sender account 2. Money is added to receiver account Both operations must succeed together.

## Transaction Flow

```
START TRANSACTION
  ↓
SQL Statements
  ↓
COMMIT / ROLLBACK
```

## Create Accounts Table

```
CREATE TABLE Accounts (
    account_id INT PRIMARY KEY,
    account_holder VARCHAR(100),
    balance DECIMAL(10,2)
);
```

## Insert Sample Data

```
INSERT INTO Accounts VALUES
(101, 'Rahul Sharma', 50000),
(102, 'Anita Verma', 30000),
(103, 'Karan Singh', 45000);
```

| account_id | account_holder | balance |
|------------|----------------|---------|
| 101        | Rahul Sharma   | 50000   |
| 102        | Anita Verma    | 30000   |
| 103        | Karan Singh    | 45000   |

## 1. COMMIT in MySQL

COMMIT permanently saves all transaction changes into the database. Once COMMIT is executed, changes cannot be undone.

## Hands-On Example: Money Transfer

```
START TRANSACTION;

UPDATE Accounts
SET balance = balance - 5000
WHERE account_id = 101;

UPDATE Accounts
SET balance = balance + 5000
WHERE account_id = 102;

COMMIT;
```

| account_holder | balance |
|----------------|---------|
| Rahul Sharma   | 45000   |
| Anita Verma    | 35000   |

- Changes are permanently saved
- Used after successful transactions
- Important in banking systems

## 2. ROLLBACK in MySQL

ROLLBACK cancels all changes made during the current transaction. Used when errors or failures occur.

```
START TRANSACTION;
```

```
UPDATE Accounts
SET balance = balance - 10000
WHERE account_id = 101;
```

```
ROLLBACK;
```

| account_holder | balance |
|----------------|---------|
| Rahul Sharma   | 50000   |

- Undo all transaction changes
- Used during failures
- Protects database consistency

## 3. SAVEPOINT in MySQL

SAVEPOINT creates checkpoints inside a transaction. Allows partial rollback instead of cancelling the complete transaction.

```
START TRANSACTION;
```

```
UPDATE Accounts
SET balance = balance - 2000
WHERE account_id = 101;
```

```
SAVEPOINT before_second_update;
```

```
UPDATE Accounts
SET balance = balance + 2000
WHERE account_id = 102;
```

```
ROLLBACK TO before_second_update;
```

```
COMMIT;
```

- Allows partial rollback
- Useful in large transactions
- Improves transaction control

## COMMIT vs ROLLBACK

| COMMIT                    | ROLLBACK                |
|---------------------------|-------------------------|
| Saves changes permanently | Undoes changes          |
| Used after success        | Used after failure      |
| Data becomes final        | Restores previous state |

## Hands-On Practice Exercises

- Create EmployeeSalary table
- Insert sample salary records
- Perform salary updates using COMMIT
- Use ROLLBACK after incorrect updates
- Use SAVEPOINT for partial rollback

## Mini Project - Banking Transaction System

- Create Accounts table
- Insert account balances
- Perform money transfers
- Use COMMIT for successful transactions
- Use ROLLBACK on failures
- Use SAVEPOINT for partial rollback
- Generate transaction reports

## Common MySQL Transaction Mistakes

- Forgetting COMMIT
- Using ROLLBACK after COMMIT
- Long-running transactions
- Not using transactions in critical systems

## MySQL Interview Questions

- What is a transaction in MySQL?
- Difference between COMMIT and ROLLBACK?
- What is SAVEPOINT?
- Can ROLLBACK undo committed changes?
- Why are transactions important?

## Final Summary

| Command   | Purpose                     |
|-----------|-----------------------------|
| COMMIT    | Save changes permanently    |
| ROLLBACK  | Undo changes                |
| SAVEPOINT | Partial rollback checkpoint |